

FLM: Forward Eligibility and Learned Choice

A controlled learning-rule comparison and physical action assay

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Abstract

We compare full backpropagation through time, fixed-core readout learning, local forward eligibility, a no-history control, and reward-modulated eligibility in a 256-neuron fly-wired rate network. Fifteen fixed-budget runs learn, reverse and restore a delayed cue association. All methods receive identical stimuli and initialization within each of three seeds. At the final 48-frame delay probe, mean accuracy is 100% for backpropagation, 86.46% for the fixed-core control, 83.33% for supervised eligibility, and 50% for both no-history and reward eligibility. Thus this experiment provides no local-learning advantage over backpropagation. A separate 40-case MuJoCo assay verifies that recorded learned choices drive physical turns through a designed gait controller. These results establish a reproducible learning-and-action interface, with substantial limitations; they do not establish language-to-motor transfer or biological validity.

1 Relation to the language model

This study is separate from FLM’s 600k-parameter WikiText and BabyLM language experiments. It reuses the fast/slow rate recurrence and anatomy-only selection procedure, with a four-channel sensory input and a two-action output. The graph contains 256 central intrinsic neurons, 6,678 directed edges and 32 readout pools. Its source is the same verified MaleCNS representation [2], but it is a different induced subset from the 1,024-neuron language core. Input channels and action projections are engineered; no sensory or descending biological pathway is claimed.

Eligibility-based learning is established prior work, applicable beyond fly wiring. The e-prop framework motivates a factorization into forward eligibility and neuron-specific learning signals [1]. Here we derive a local temporal approximation for FLM’s particular normalized rate dynamics. The model has no spikes, synaptic delays or biophysical units, and is not a reproduction of e-prop’s spiking experiments.

2 Dynamics and direct derivatives

For edge $j \rightarrow i$, let $v_{ij} = b_{ij} \exp(\theta_{ij})$ and $W_{ij} = v_{ij} / \sum_k |v_{ik}|$, where b_{ij} carries the retained contact-derived magnitude and assumed presynaptic sign. Edge logits are projected to $[-3, 3]$ after updates. With $q_i = \sum_j W_{ij} h_j^{\text{prev}}$, the network uses

$$z_i = \tanh(A_i x + b_i + g q_i), \quad (1)$$

$$h_i = (1 - \alpha_i) h_i^{\text{prev}} + \alpha_i z_i, \quad (2)$$

$$s_i = (1 - \beta_i) s_i^{\text{prev}} + \beta_i h_i. \quad (3)$$

The existing sigmoid bounds give $\alpha \in (.05, .95)$, $\beta \in (.002, .100)$, and $g \in (.05, 3)$. The normalized edge’s direct derivative is

$$\frac{\partial q_i}{\partial \theta_{ij}} = W_{ij} h_j^{\text{prev}} - |W_{ij}| q_i. \quad (4)$$

Ignoring the second term would change the learning rule: incoming normalization couples the effects of all weights arriving at the same postsynaptic neuron.

3 Forward credit and controls

Define the local fast-state Jacobian $J_i = 1 - \alpha_i + \alpha_i(1 - z_i^2)gW_{ii}$. For a parameter associated with neuron i , retain fast and slow traces:

$$e_{h,i} \leftarrow J_i e_{h,i}^{\text{prev}} + d_{h,i}, \quad (5)$$

$$e_{s,i} \leftarrow (1 - \beta_i)e_{s,i}^{\text{prev}} + \beta_i e_{h,i} + d_{\beta,i}. \quad (6)$$

Direct input derivatives use $\alpha_i(1 - z_i^2)x$; bias replaces x with one. Edge derivatives multiply equation (4) by $\alpha_i(1 - z_i^2)g$. Alpha and gain logits use their direct sigmoid derivatives. Beta contributes $d_{\beta,i} = (h_i - s_i^{\text{prev}})\partial\beta_i/\partial\text{logit}$. The complete expressions and implementation are in the declared protocol.

At the final decision, instantaneous derivatives through pooling, LayerNorm and the readout supply L_h, L_s . Core updates use $\sum_{\text{batch}}(L_h e_h + L_s e_s)$; the shared gain also sums over neurons. Earlier states have no autograd history in the forward conditions. This approximation drops temporal credit carried through other neurons. Tests match full autograd when those routes are absent, verify the normalized direct derivative by finite differences, and check exact state parity between the ordinary and eligibility paths. They do not prove equality with the full recurrent gradient.

All conditions use SGD at .03, batch eight, no momentum or weight decay, and global gradient clipping at one. The fixed-core control trains only 258 output and normalization parameters. The other conditions may train all 8,729 parameters. The no-history control clears trace history each frame. The reward condition samples an action and uses $-(r - \bar{r}) \log p(a)$, with $r \in \{0, 1\}$ and the previous exponential reward baseline (decay .95, initially .5). Its separate action RNG preserves the shared sensory stream. Binary correctness also reveals whether the other action would succeed, so this task does not demonstrate learning from strictly less target information.

4 Registered task and final measurements

Two cue frames precede an integer delay sampled uniformly from 4–12 and one query frame. An independent Gaussian distractor (SD .2) is present throughout. Each balanced batch contains four examples of each cue. The original association is trained for 300 updates, its reverse for 300, then the original for 300: 7,200 episodes per run. Seeds are 17, 29 and 41. State and traces reset each episode; weights persist. No phase flag is supplied.

Every 100 updates and initially, fixed panels contain 256 balanced episodes per delay (4, 8, 12, 24, 48). Panels are repeatedly inspected diagnostics, not an untouched test set or a checkpoint-selection criterion. Table 1 reports the restored rule after the fixed 900 updates. The two association scores are mutually exclusive, not independent tasks.

Table 1: Final accuracy (%). Means over three seeds; the last column is the observed seed range at delay 48, not a confidence interval.

Learning rule	Delay 8	Delay 12	Delay 48	Range, 48
BPTT	100.00	100.00	100.00	100.0–100.0
Fixed core	100.00	99.74	86.46	74.2–100.0
Eligibility	100.00	85.29	83.33	50.0–100.0
No trace history	88.28	66.67	50.00	50.0–50.0
Reward + eligibility	83.33	83.33	50.00	50.0–50.0

5 Reversal and retention

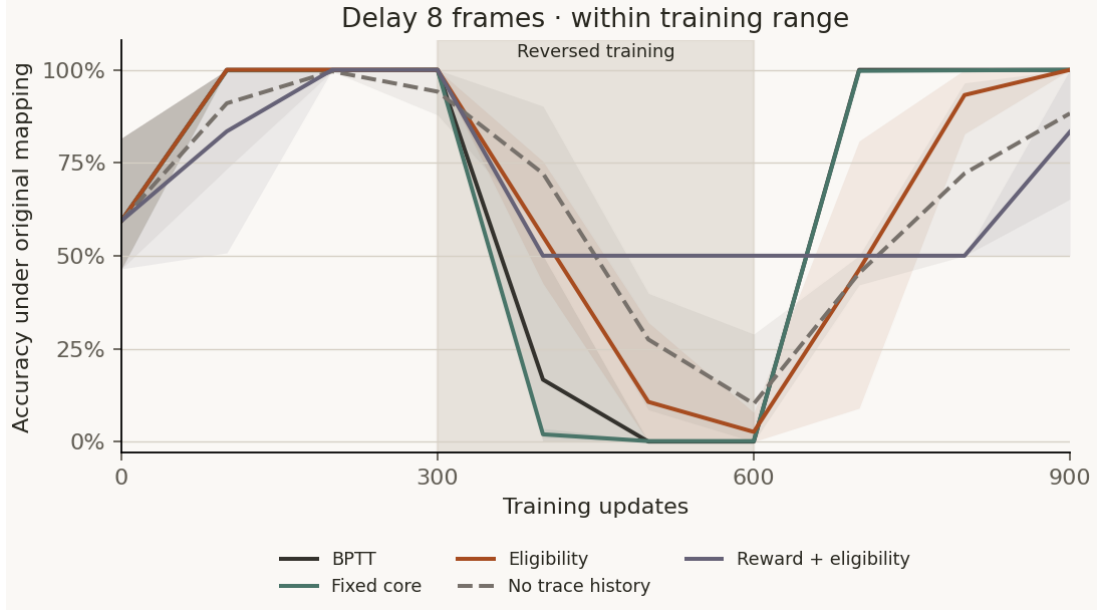


Figure 1: Within-range delay. The vertical axis always scores the original association; descending during reversed training therefore indicates learning the new rule. Curves average three seeds; bands span their observed range.

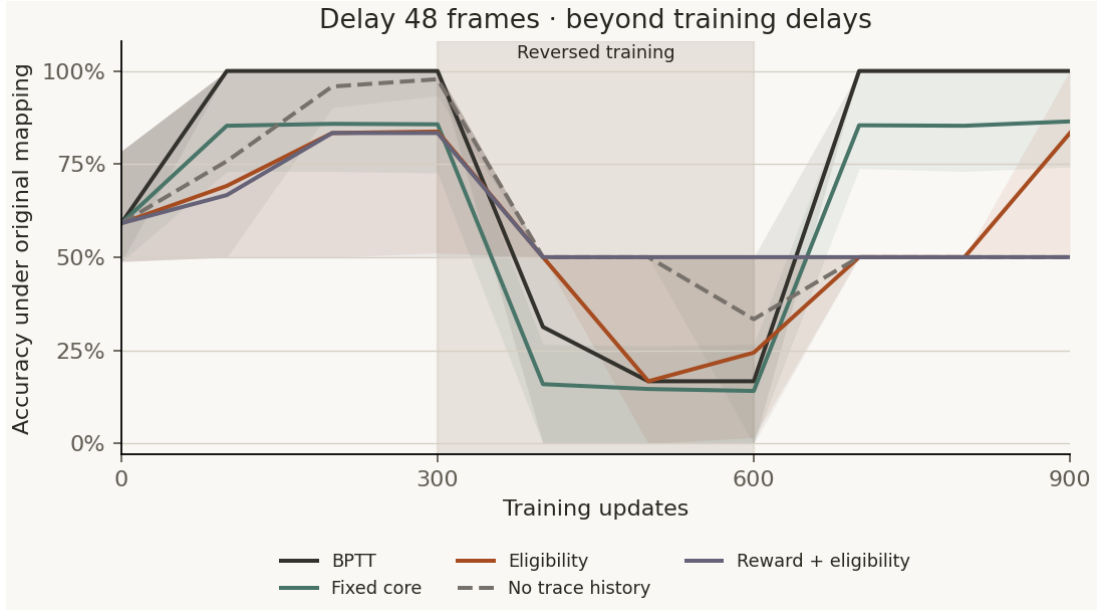


Figure 2: Delay extrapolation. Backpropagation finishes at 100% in all three seeds. Supervised eligibility finishes at 50%, 100%, 100%; the fixed core finishes at 100%, 85.16%, 74.22%. No-history and reward eligibility finish at 50% in every seed. This narrow task does not establish a general ranking.

6 From a learned choice to a physical turn

The predeclared physical assay uses seed 17, checkpoints 0/300/600/900 and both zero-distractor cues at delay eight, for each of the five learning conditions. Argmax action zero selects descending command [4, 1.2]; action one selects [1.2, 4]. Each case starts independently from the same body pose and controller seed, and runs one simulated second (10,000 physics steps) using FlyGym 2.1.0, MuJoCo 3.9.0 and its designed hybrid turning controller [3].

The complete assay contains 40 separately simulated cases and 2 unique descending commands. The measured heading sign matches the chosen command in all 40 cases. Neural choices agree with the rule associated with their checkpoint in 33 of 40 cases, including the untrained controls. All recorded generalized positions are exactly equal between cases with the same command and initialization. This confirms deterministic action execution, not independent motor skills or a statistical motor-learning benchmark.

The two commands change heading by approximately +2.811 and −2.556 radians, respectively. Training changes cue-dependent action probabilities and can change the resulting trajectory. The published archive binds each neural decision and state to its checkpoint and physical trajectory. The video shows the predeclared supervised-eligibility, update-300, cue-zero case.

The neural decision is recorded before the physics replay. FLM does not receive online proprioception, train joint trajectories, learn balance or adapt from a physical reward in this assay. The gait and contact controller supply those motor functions. The female NeuroMechFly body accompanies a male-connectome subset. These limits are distinct from the browser body’s illustrative pose and from any future transfer of language-trained state into motor decisions.

7 What the experiment changes

The implemented forward rule is executable and can learn a cue association, but it is not a demonstrated improvement over full backpropagation. The fixed-core control’s strength exposes a limitation of the task: accessible state already contains much of the required information. Reversal and long-delay failures show sensitivity to plasticity and initialization, not proof of a biological mechanism. Anatomical advantage needs a matched rewired graph; useful learned memory needs tasks that a fixed readout cannot readily solve.

Eligibility storage is 558,464 tensor bytes for batch eight, plus 16,384 recurrent state bytes. It is fixed with episode length but grows with parameter count. This is not a measured peak-memory, runtime or energy advantage over BPTT. The output normalization and feedback routes are engineered and nonlocal. Next comparisons should isolate plasticity sites, inspect gradient alignment, strengthen the memory task and add sensory feedback. Language transfer and scaling remain unfinished experiments.

8 Reproduction and artifacts

Run `python -m flm.behavior_study`, then `python -m flm.behavior_report`. Use the isolated embodied environment for `python experiments/embodiment/learned_choice.py -video`, followed by `python scripts/physical_choice_report.py`. Source revisions, stimulus stream hashes, every panel prediction, failed cases and neural states are retained. The website publishes JSON, CSV, plotted data and trajectory archives. The implementation and larger program are available at <https://github.com/Kuberwastaken/flm> and <https://flm.kuber.studio/#research>.

References

- [1] Guillaume Bellec et al. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 2020. <https://doi.org/10.1038/s41467-020-17236-y>.
- [2] Janelia Research Campus. MaleCNS connectome release and downloads. <https://male-cns.janelia.org/download/>, 2026.
- [3] S. Wang-Chen et al. NeuroMechFly v2: simulating embodied sensorimotor control in adult *Drosophila*. *Nature Methods*, 21:2353–2362, 2024. <https://doi.org/10.1038/s41592-024-02497-y>.